

1. Let $U_1, U_2, \dots$ be an i.i.d. sequence of $U[0, 1]$ random variables and let X be a Lévy process independent of $U_1, U_2, \dots$, and let $\bar{X}_t = \sup_{0 \leq s \leq t} X_s$. From the lemma below

$$\bar{X}_t \stackrel{(d)}{=} (X_{tU_1}^{(1)})^+ + \bar{X}_{t(1-U_1)} \quad (1)$$

where $X^{(1)}$ is an independent copy of X . Iterate this relation to show how we can sample $\bar{X}_t$, and comment on the truncation error if X is a square integrable martingale.

Solution. Using the tower property and (1) we see that

$$\mathbb{E}(1_{\bar{X}_{t(1-U_1)} \leq x}) = \mathbb{E}(\mathbb{E}(1_{\bar{X}_{t(1-U_1)} \leq x} | U_1)) = \mathbb{E}(\mathbb{E}(1_{(X_{t(1-U_1)U_2}^{(2)})^+ + \bar{X}_{t(1-U_1)(1-U_2)} \leq x} | U_1)) = \mathbb{E}(1_{(X_{t(1-U_1)U_2}^{(2)})^+ + \bar{X}_{t(1-U_1)(1-U_2)} \leq x})$$

where $X^{(2)}$ is another independent copy of X , so $\bar{X}_{t(1-U_1)}$ and $(X_{t(1-U_1)U_2}^{(2)})^+ + \bar{X}_{t(1-U_1)(1-U_2)}$ have the same cdf (and hence distribution).

Then

$$\begin{aligned} \bar{X}_t &\stackrel{(d)}{=} (X_{tU_1}^{(1)})^+ + (X_{t(1-U_1)U_2}^{(2)})^+ + \bar{X}_{t(1-U_1)(1-U_2)} \\ &\stackrel{(d)}{=} (X_{tU_1}^{(1)})^+ + (X_{t(1-U_1)U_2}^{(2)})^+ + (X_{tU_3(1-U_1)(1-U_2)}^{(3)})^+ + \bar{X}_{t(1-U_1)(1-U_2)(1-U_3)} \\ &\dots \\ &\stackrel{(d)}{=} \sum_{n=1}^N \xi_n^+ + \bar{X}_{t \prod_{i=1}^N (1-U_i)} \end{aligned} \quad (2)$$

where $\xi_n = X_{t \prod_{i=1}^{n-1} (1-U_i)}^{(n)}$ and $X^{(1)}, X^{(2)}, \dots$ are i.i.d. copies of the original X process. The left sum here is increasing in N , and the final term is decreasing in N , so they both converge to a limit a.s., but we need to verify that the second limit is zero.

To this end, let $\bar{U}_i = 1 - U_i$. Then the $\bar{U}_i$'s are also i.i.d. $U[0, 1]$, and $-\log(\prod_{i=1}^N (1 - U_i)) = -\sum_{i=1}^N \log \bar{U}_i \rightarrow \infty$ a.s. as $N \rightarrow \infty$ (because $-\frac{1}{N} \sum_{i=1}^N \log \bar{U}_i \rightarrow -\mathbb{E}(\log \bar{U}_i) = 1$ from the SLLN) so the $\prod_{i=1}^N (1 - U_i)$ term in (2) tends to $e^{-\infty} = 0$, and hence so does $\bar{X}_{t \prod_{i=1}^N (1-U_i)}$, because right continuity of X implies that for all $\varepsilon > 0$ there exists a $\delta = \delta(\omega) > 0$ such that $|X_t| \leq \varepsilon$ for all $t \in [0, \delta]$, hence $\bar{X}_t \leq \varepsilon$ for all $t \in [0, \delta]$, so $\bar{X}_t \searrow 0$ a.s. as $t \searrow 0$.

We have taken the limit as $N \rightarrow \infty$ in (2) and shown that the remainder term tends to zero, thus

$$\bar{X}_t \stackrel{(d)}{=} \sum_{n=1}^{\infty} \xi_n^+ \quad (3)$$

which gives us a method for sampling $\bar{X}_t$ in terms of increments of X at uniform times. This is known as the **stick-breaking algorithm** (see remark below for further explanation on this). If X is an α -stable process, recall we can efficiently sample increments of X using the Chambers, Mallows & Stuck method from q8 in Hwk3.

Remark 0.1 Similarly, we can jointly sample $(X_t, \bar{X}_t, \tau_t)$ as

$$(X_t, \bar{X}_t, \tau_t) \stackrel{(d)}{=} \sum_{n=1}^{\infty} (\xi_n, \xi_n^+, \ell_n 1_{\xi_n > 0})$$

where $\tau_t = \inf\{s : \bar{X}_s = \bar{X}_t\}$ and $\ell_n = U_n \prod_{i=1}^{n-1} (1 - U_i)$, but we won't prove this here.

To analyze the error incurred in truncating Eq (3) after N terms, we set $\varepsilon_N := \prod_{i=1}^N (1 - U_i)$ (i.e. the product term in Eq (2)); then

$$\begin{aligned} \mathbb{E}[\varepsilon_N] &= \mathbb{E}[(1 - U_1)]^N = \left(\int_0^1 (1 - u) du\right)^N = \left(\frac{1}{2}\right)^N \\ \mathbb{E}[\varepsilon_N^2] &= (\mathbb{E}[(1 - U_1)^2])^N = \left(\int_0^1 (1 - u)^2 du\right)^N = \left(\frac{1}{3}\right)^N \end{aligned}$$

If we assume X is a square integrable martingale so $\mathbb{E}(e^{iuX_t}) = e^{t\psi(u)}$ then $\mathbb{E}(X_t^2) = -\frac{d^2}{du^2} \mathbb{E}(e^{iuX_t})|_{u=0} = -\psi''(0)t < \infty$ (since $\psi'(0) = 0$ because $\mathbb{E}(X_t) = 0$ from the martingale assumption); then from the Doob maximal inequality for submartingales (generalization of q2 below) and the tower property we see that

$$\begin{aligned} \mathbb{E}(\bar{X}_{t \prod_{i=1}^N (1-U_i)}^2) &= \mathbb{E}(\mathbb{E}(\bar{X}_{t \prod_{i=1}^N (1-U_i)}^2 | U_1, U_2, \dots)) \leq 4\mathbb{E}(\mathbb{E}(X_{t \prod_{i=1}^N (1-U_i)}^2 | U_1, U_2, \dots)) \\ &= 4\mathbb{E}(X_{t \prod_{i=1}^N (1-U_i)}^2) \\ &= 4|\psi''(0)|t \mathbb{E}\left(\prod_{i=1}^N (1 - U_i)\right) = \frac{\text{const.}}{2^N} \end{aligned}$$

so the truncation error $\bar{X}_t^2 \prod_{i=1}^N (1-U_i)$ tends to zero in L^2 geometrically fast (and hence also in probability).

Remark 0.2 We can regard $\ell_1 = U_1$, $\ell_2 = U_2(1-U_1)$, $\ell_3 = U_3(1-U_1)(1-U_2)$, ... as the **lengths of a stick** which is successively broken at a uniform point (with all breaks independent).

See short piece of code to compute (3) at <https://colab.research.google.com/drive/1owlTCYoUNgZ03ghUcM54ac0JG06i-ivw> for standard Brownian motion, which also checks that the sampled $\bar{X}_t$ has the correct distribution, which we know from the reflection principle.

Claim: $\bar{X}_{e_q} \stackrel{(d)}{=} \bar{X}_{e_q U} + \tilde{X}_{e_q(1-U)}^+$, and hence $\bar{X}_t \stackrel{(d)}{=} \bar{X}_{tU} + \tilde{X}_{t(1-U)}^+$

Lemma 0.1

$$\mathbb{E}(e^{iz\bar{X}_{e_q}}) = \mathbb{E}(e^{iz(\bar{X}_{e_q U} + \tilde{X}_{e_q(1-U)}^+)})$$

where $\tilde{X}$ is an independent copy of X .

Proof. Set $S := e_q U$, $T := e_q(1-U)$. By comparing joint mgfs, we can check that (S, T) has joint density

$$\mathbb{P}(S \in ds, T \in dt) = \frac{qe^{-q(s+t)}}{s+t} ds dt, \quad s, t > 0.$$

Now define

$$\varphi_t(z) := \mathbb{E}(e^{iz\bar{X}_t}), \quad m_t(z) := \mathbb{E}(e^{izX_t^+}).$$

Since $\tilde{X}_t$ and X_t are independent for t fixed, we see that

$$\mathbb{E}(e^{iz(\bar{X}_{e_q U} + \tilde{X}_{e_q(1-U)}^+)}) = \int_0^\infty \int_0^\infty \frac{qe^{-q(s+t)}}{s+t} \varphi_s(z) m_t(z) ds dt. \quad (4)$$

Using $\frac{1}{s+t} = \int_0^\infty e^{-r(s+t)} dr$, and Fubini's theorem,

$$\mathbb{E}(e^{iz(\bar{X}_{e_q U} + \tilde{X}_{e_q(1-U)}^+)}) = q \int_0^\infty \left(\int_0^\infty e^{-(q+r)s} \varphi_s(z) ds \right) \left(\int_0^\infty e^{-(q+r)t} m_t(z) dt \right) dr.$$

Recall that

$$\Phi_\lambda^+(z) = \mathbb{E}(e^{iz\bar{X}_{e_\lambda}}) = \lambda \int_0^\infty e^{-\lambda s} \varphi_s(z) ds.$$

Likewise if $M_\lambda(z) := \mathbb{E}(e^{izX_{e_\lambda}^+})$, then $\int_0^\infty e^{-\lambda t} m_t(z) dt = \frac{M_\lambda(z)}{\lambda}$. Hence (making the change of variable $\lambda = q + r$ in (4)), we see that

$$\mathbb{E}(e^{iz(\bar{X}_{e_q U} + \tilde{X}_{e_q(1-U)}^+)}) = q \int_q^\infty \Phi_\lambda^+(z) M_\lambda(z) \frac{1}{\lambda^2} d\lambda. \quad (5)$$

Now use the Fristedt representation from Hwk2:

$$\Phi_\lambda^+(z) = \exp\left(\int_0^\infty \frac{e^{-\lambda t}}{t} \int_0^\infty (e^{izx} - 1) \rho_t(x) dx dt\right).$$

Since

$$m_t(z) - 1 = \int_0^\infty (e^{izx} - 1) \rho_t(x) dx,$$

this may be rewritten as

$$\Phi_\lambda^+(z) = \exp\left(\int_0^\infty \frac{e^{-\lambda t}}{t} (m_t(z) - 1) dt\right).$$

Differentiating,

$$\partial_\lambda \log \Phi_\lambda^+(z) = - \int_0^\infty e^{-\lambda t} (m_t(z) - 1) dt.$$

Hence

$$M_\lambda(z) = 1 + \lambda \int_0^\infty e^{-\lambda t} (m_t(z) - 1) dt = 1 - \lambda \partial_\lambda \log \Phi_\lambda^+(z) = 1 - \lambda \frac{\partial_\lambda \Phi_\lambda^+(z)}{\Phi_\lambda^+(z)}.$$

It follows that

$$\frac{\Phi_\lambda^+(z)M_\lambda(z)}{\lambda^2} = \frac{\Phi_\lambda^+(z)}{\lambda^2} - \frac{\partial_\lambda \Phi_\lambda^+(z)}{\lambda} = -\partial_\lambda \left(\frac{\Phi_\lambda^+(z)}{\lambda} \right).$$

Substituting this into (5)

$$\mathbb{E}(e^{iz(\bar{X}_{e_q}U + \bar{X}_{e_q(1-U)}^+)}) = q \int_q^\infty -\partial_\lambda \left(\frac{\Phi_\lambda^+(z)}{\lambda} \right) d\lambda.$$

Therefore,

$$\mathbb{E}(e^{iz(\bar{X}_{e_q}U + \bar{X}_{e_q(1-U)}^+)}) = q \left(\frac{\Phi_q^+(z)}{q} - \lim_{\lambda \rightarrow \infty} \frac{\Phi_\lambda^+(z)}{\lambda} \right).$$

Since $e_\lambda \rightarrow 0$ in probability as $\lambda \rightarrow \infty$, $\bar{X}_{e_\lambda} \rightarrow 0$ in probability, so

$$\Phi_\lambda^+(z) \rightarrow 1, \quad \frac{\Phi_\lambda^+(z)}{\lambda} \rightarrow 0.$$

Thus $\mathbb{E}(e^{iz(\bar{X}_{e_q}U + \bar{X}_{e_q(1-U)}^+)}) = \Phi_q^+(z) = \mathbb{E}(e^{iz\bar{X}_{e_q}})$. as claimed. ■

1. Let $U_1, U_2, \dots$ and $Z_1, Z_2, \dots$ be an i.i.d. sequence of $U[0, 1]$ and $N(0, 1)$ random variables respectively (also independent of each other) and let $M = \bar{W}_1$, where $\bar{W}_t = \max_{0 \leq s \leq t} W_s$ and W is a Brownian motion independent of both sequences. It can be shown that

$$\bar{X}_1 \stackrel{(d)}{=} (X_{tU_1}^1)^+ + \bar{X}_{t(1-U_1)}.$$

By iterating this relation, derive the stick-breaking algorithm for sampling M . and characterize the speed of convergence.

$$\begin{aligned} \bar{X}_t &\stackrel{(d)}{=} (X_{tU_1}^1)^+ + (X_{t(1-U_1)U_2}^2)^+ + \bar{X}_{t(1-U_1)(1-U_2)} \\ \dots &\stackrel{(d)}{=} \sum_{n=1}^N (X_{tU_n}^{(n)})^+_{\prod_{i=1}^{n-1}(1-U_i)} + \bar{X}_{t\prod_{i=1}^N(1-U_i)} \end{aligned}$$

Let $U_1, U_2, \dots$ and $Z_1, Z_2, \dots$ be an i.i.d. sequence of $U[0, 1]$ and α -stable random variables respectively (also independent of each other) and let X be an α -stable process with $\alpha \in (0, 2]$ independent of both sequences, and you may assume that $X_1 \stackrel{(d)}{=} Z_i$ for $i = 1, 2, \dots$. It can be shown that

$$\bar{X}_1 \stackrel{(d)}{=} U_1^{\frac{1}{\alpha}} Z_1^+ + (1 - U_1)^{\frac{1}{\alpha}} \bar{X}_1$$

where $\bar{X}_t = \sup_{0 \leq s \leq t} X_s$. Iterate this relation to show how we can sample $\bar{X}_1$ in terms of $Z_1^+, Z_2^+, \dots$, and comment on the error.

Solution.

$$\begin{aligned} \bar{X}_1 &\stackrel{(d)}{=} \sqrt{U_1} Z_1^+ + \sqrt{1 - U_1} (\sqrt{U_2} Z_2^+ + \sqrt{1 - U_2} \bar{X}_1) \\ &\stackrel{(d)}{=} \sqrt{U_1} Z_1^+ + \sqrt{1 - U_1} (\sqrt{U_2} Z_2^+ + \sqrt{1 - U_2} (\sqrt{U_3} Z_3^+ + \sqrt{1 - U_3} \bar{X}_1)) \\ \dots &\stackrel{(d)}{=} \sum_{n=1}^N Z_n^+ \sqrt{U_n} \prod_{i=1}^{n-1} \sqrt{1 - U_i} + \bar{X}_1 \prod_{i=1}^N \sqrt{1 - U_i} \end{aligned}$$

where Z_i 's are i.i.d. The sum here is increasing in N , and the 2nd term is decreasing in N , so they both converge a.s.

Now let $\bar{U}_i = 1 - U_i$. Then $\bar{U}_i$ is also $U[0, 1]$, and $-\log(\prod_{i=1}^N \sqrt{1 - U_i}) = -\frac{1}{2} \sum_{i=1}^N \log \bar{U}_i$ is the sum of N i.i.d. $\text{Exp}(1)$ random variables, which tend to infinity, so $\prod_{i=1}^N \sqrt{1 - U_i}$ tends to $e^{-\infty} = 0$. Hence we can sample $\bar{X}_1$ as

$$\sum_{n=1}^{\infty} Z_n^+ \sqrt{U_n} \prod_{i=1}^{n-1} \sqrt{1 - U_i}$$

This is known as the **stick-breaking algorithm**.

For the error, we first note that $-\frac{1}{N} \sum_{i=1}^N \log \bar{U}_i \sim \varepsilon_N$ where ε_N tends weakly to a $\mathcal{N}(1, \frac{1}{N})$ random variable as $N \gg 1$ from the CLT, so the error is

$$M e^{-\frac{1}{2} N \varepsilon_N}$$

for $N \gg 1$, and the exponential term here has mean $e^{-\frac{3N}{8}}$ and variance $e^{-\frac{1}{2}N} - e^{-\frac{3}{4}N}$, so the bias and sample variance are exponentially small.

If X is a symmetric α -stable process, we just modify *** to

$$M \stackrel{(d)}{=} U^{1/\alpha} Z_1^+ + (1 - U)^{1/\alpha} M.$$

and continue in the same way, although in this case $\mathbb{E}(M^2) = \infty$.

Stick breaking $E \sim -\log U$, so $U = e^{-E}$

$$\mathbb{E}(e^{pZ_n^+ \prod_{i=1}^n U_i}) = \mathbb{E}(2^n \sum_{k=0}^{\infty} \frac{(pZ_n^+)^k}{k!(k+2)^n})$$